Basic Numerical Analysis

Syllabus

Autumn 2026 **Math 21100**

§A. What is this class?

Math 21100 is an introduction to numerical analysis. Most problems of continuous mathematics (involving real or complex variables as opposed to integers), such as finding roots of a function, evaluating an integral, or solving a differential equation, cannot be answered exactly by any finite computation, even in perfect arithmetic. In practice, we solve them with algorithms that produce approximations. We often prefer an approximation even when an exact finite procedure exists. Gaussian elimination solves a linear system in finitely many steps, for instance, but for a large system an iterative method reaches a good approximation much sooner. But an approximate answer raises some questions. How close is the computed answer to the true one? How quickly does the error decrease as we refine the computation, and can further refinement ever make the answer less accurate? How can we tell, from the computation alone, whether the answer should be trusted?

Contrary to a common characterization, numerical analysis is not the study of rounding errors. Lloyd N. Trefethen's definition describes the subject more accurately.

Numerical analysis is the study of algorithms for the problems of continuous mathematics.

The course is organized around why different numerical methods work and when they fail, rather than around a catalog of recipes. For each method, we ask what the hypotheses are, how fast it converges, and what to do when the hypotheses are not satisfied. We then test the theory against computed approximations and explain any disagreement. By the end of the quarter, you should be able to answer the three questions in the first paragraph for every method we study.

A.1 What you need coming in

This is an upper-level applied math course. You will be most successful if you are comfortable with the following at minimum:

- differential and integral calculus, including the mean value theorem, Rolle's theorem, the intermediate value theorem, uniform continuity on a closed bounded interval, and Taylor's theorem with the Lagrange remainder,
- reading and writing a rigorous proof (at the level of an undergraduate analysis sequence),
- basic multivariable calculus (vectors, partial derivatives, the Jacobian matrix), and
- basic linear algebra (multiplying and inverting matrices and calculating eigenvalues).

Prior programming experience is helpful but not necessary. In-class demonstrations are done in Python, using Jupyter notebooks in VS Code. See section E for AI policies.

A.2 How class meetings work

I will post lecture notes on Canvas before each meeting, so you can read along and not spend valuable class time transcribing board work. The notes are written to promote **discovery-based active learning**, and I plan to incorporate as much of that as possible during our meetings. This is *not* a student-led inquiry-based learning course (I will lecture when it is the right tool), but even during lectures, you should expect to be actively involved, supplying key steps in proofs and testing ideas before they are formalized. Come prepared to think, write, and talk with your classmates. For some meetings, I might ask you to bring your laptop or tablet so that you can run some computation yourself during class.

Please make use of my office hours and plan to work hard. Homework and office hours are the place to make mistakes and learn from them, and the assessments reward that preparation.

§B. Key information

Class meetings

MWF, 12:30–1:20 pm, Eckhart 202

How to contact me

Email: subhadip@uchicago.edu
Office: Eckhart 120B

Office hours

See Canvas for up-to-date hours. You can also email me to set up an individual meeting.

Study materials

- **Lecture notes:** The notes are written for this course and are its primary resource. **No textbook purchase is required.**
- **References (recommended, not required):**
  - Süli and Mayers, *An Introduction to Numerical Analysis*, Cambridge University Press, 2003. ISBN 978-0-521-00794-8.
  - Scott, *Numerical Analysis*, second edition, 2016. Free PDF on the author's website.
  - Burden, Faires, and Burden, *Numerical Analysis*, 10th edition, Cengage, 2016. ISBN 978-1-305-25366-7.
- **Supplementary reading:** L. N. Trefethen, "The Definition of Numerical Analysis," *SIAM News*, November 1992, and his article "Numerical Analysis" in *The Princeton Companion to Mathematics* (2008).
- **Computing:** I will use VS Code with an offline installation of Python. All code for the course is written in Python, and you may run it in any environment you like, such as VS Code, JupyterLab, or Google Colab. Code for computational problems is submitted as a Jupyter notebook (`.ipynb` file), as described in section F.2.

Class announcements

Available on: `https://canvas.uchicago.edu/courses/73947`
Check Canvas and your UChicago email at least once before and after each class.

§C. Topics covered

Over the next nine weeks, we will cover a **subset** of the topics listed below. The course standards that will show up in timed assessments will be released on Canvas in phases during the quarter.

(1) Interpolation. Existence and uniqueness, Lagrange form, the interpolation error theorem, Runge's phenomenon and Chebyshev nodes, piecewise polynomials and splines.

(2) Numerical differentiation. Difference formulas from the interpolant, truncation error against rounding error, the optimal step size, Richardson extrapolation.

(3) Approximation theory. Norms on function spaces, best approximation, inner products and orthogonality, orthogonal polynomials and the three-term recurrence, least squares.

(4) Quadrature. Interpolatory rules, degree of exactness and error bounds, Newton–Cotes and composite rules, Gauss quadrature, adaptive schemes.

(5) Iteration and linear systems. Elimination algorithms and pivoting, vector and matrix norms, conditioning, Jacobi and Gauss–Seidel from a splitting, convergence criteria and the spectral radius.

(6) Nonlinear equations. Bisection, fixed-point iteration and contraction, Newton and secant methods, order of convergence, Newton in $\mathbb{R}^n$.

(7) Initial-value problems for ODEs. Euler's method, local truncation error against global error, consistency and convergence, Runge–Kutta methods, predictor-corrector methods.

§D. Learning objectives

Upon completion of this course, you will be able to

- **state** the definitions and theorems that establish when a method works and how accurate it is;
- **derive** an error bound for a method, **identify** the hypothesis each step of the derivation uses, and **explain** what fails when a hypothesis is removed;
- **predict** the rate at which an algorithm converges as a step size shrinks or a node count grows, and **verify** the prediction against computed output;
- **implement** an algorithm in Python, **test** it on a case with a known answer, and **interpret** its output using the theorems of the course;
- **diagnose** computed results that disagree with the theory, including results from someone else's code, and **identify** the mechanism that produced them;
- **distinguish** the sensitivity of an ill-conditioned problem from the loss of accuracy an algorithm introduces, and **explain** which of the two a better algorithm can repair;
- **select** a method for a problem with stated constraints on data, smoothness, and cost, **justify** the choice with a result from the course, and **check** that the result's hypotheses hold for the problem.

§E. Using GenAI in Math 21100

Code. You may use generative AI tools (such as ChatGPT, Claude, Gemini, or PhoenixAI) to write, change, or debug the Python for any homework problem, as much as you like, on three conditions.

- Make sure the code carries out the algorithm itself. Unless otherwise noted, do not call a library function that performs the method or any part of it, such as `numpy.polyfit`, `scipy.interpolate`, `scipy.integrate`, `numpy.linalg.solve`, or `scipy.optimize`. You may use arrays, arithmetic, elementary functions, and plotting, and you may reuse any code posted on Canvas, including the library calls it makes.
- Before you use the code on the assigned problem, check it on a problem whose answer you already know, such as an integral or a root you can compute by hand. Describe that check in your report.
- Cite the tool in your report, and say which functions or cells it wrote.

Mathematics and reports. Proofs, derivations, and the written interpretation of a computation are the practice that prepares you for the assessments, so I recommend doing them without AI. If you do use AI for this work, I will not penalize it, provided you cite it clearly (the tool, what you used it for, and what you changed) and can explain everything you submit. Any number quoted in a report must come from your own run of the code.

At the oral final exam. You may be asked to explain any line of code you submitted, including lines an AI wrote, and to predict how the output would change for a different value of a parameter.

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§F. Components of the course

Our course uses a **mastery-based grading system** (also called standards-based grading). Your course grade is mainly determined by three components: how many of the course standards you have demonstrated mastery of, your overall homework score, and your oral final exam. **The timed assessments carry no point values, and there are no weighted averages and no curves**.

F.1 Attendance and engagement

Class is your best chance to discuss new material, ask questions, and clear up confusion, so please don't miss it!

Being present

Please show up to class **on time** and ready to engage. **Attendance will be taken.** If you need to miss class for any reason, make a plan to catch up. In particular, you should:

- Read through the class notes posted to Canvas;
- Review the corresponding content using a reference text or other resources;
- Try the relevant homework problems *before* seeking additional help from me with the material from that day.

Please do not miss class without contacting me. I don't care whether you think it's a good 'excuse' or not. Just send me an email or reply to mine! Any absence you tell me about, before class or as soon as possible afterward, is excused. This isn't about enforcing perfect attendance; it's about keeping an open line of communication so I can support you effectively. Lack of communication about absences may affect your final grade, as described in section G.2.

Engaging actively

We'll be learning together, and the participation of every student, whether in whole-class, small-group, or individual settings, creates the best learning environment for everyone. You can demonstrate your engagement in a number of ways:

- Working seriously on in-class exercises and computations, both individually and with your neighbors.
- Giving constructive feedback during in-class discussions.
- Asking relevant questions in class, during office hours, or through email.
- Creating an inclusive and welcoming class environment for peers.

F.2 Weekly homework

Homework problems will be assigned via Canvas **after every lecture by 6 PM** on that day. Friday, Monday, and Wednesday homework will be **due the following Monday at 11:59 PM**. Specifically,

- the homework problems assigned on Week 1 Monday and Week 1 Wednesday will be due on Week 2 Monday.
- the homework problems assigned on Week 1 Friday, Week 2 Monday, and Week 2 Wednesday will be due on Week 3 Monday.
- Week 8 Friday, Week 9 Monday, and Week 9 Wednesday homework will be due on Week 9 Friday (with up to three days of allowed extension).

Please start working on the homework as soon as it is uploaded and utilize the office hours as necessary; do not wait until the due date to begin.

Each weekly set will have theoretical and computational problems.

- **Theoretical problems** ask you to justify results we use in class, and to prove claims and corollaries.
- **Computational problems** ask you to implement an algorithm or run an experiment, and to explain the output in a short report using theory from class or from the theoretical problems.

For some computational problems I will supply a Jupyter notebook with part of the code written and the rest left for you.

Solutions are posted after the deadline. Each problem earns 0, 1, 2, or 3 points.

- 0/3 — Not attempted, substantially incomplete, or major logical errors.

- 1/3 — Substantial effort but significant gaps, or correct core idea with inadequate justification. A computational answer that reports output without explaining it is in this category.
- 2/3 — Essentially correct with minor errors* or omissions; reasoning is mostly clear. Graphs with missing labels are in this category.
- 3/3 — Correct, complete, and clearly written; arguments are well organized. The answer discusses why, not just what. The writing is in an academic tone, and graphs are well labeled.

Your homework score is a percentage. It is the total number of points you earn on all problems of all assignments, divided by 95% of the total points possible, and capped at 100%.

$$\text{homework score} = \min\left\{100\%,\ \frac{\text{points earned}}{0.95 \times \text{points possible}}\right\}.$$

So, earning 95% of the possible points gives the maximum possible homework score. The policy is to encourage learning through mistakes on homework assignments without penalizing you severely for them.

Submission requirements

Submit homework through Gradescope. Every homework will have a PDF submission. A homework with computational problems will also have a notebook submission, due at the same time. Whatever is on Gradescope at the deadline, or at the extended deadline if you use an extension, will be final.

The PDF. Put all of your solutions in the PDF, including the reports for computational problems. I expect it to be typeset in LaTeX, and an optional LaTeX template for homework will be available on Canvas. If this creates a barrier, please contact me as soon as possible. The PDF must adhere to the following format requirements:

- Use at least 11 pt font, and start each problem on a new page.
- Clearly label and number problems.
- Write in complete sentences.
- For computational problems, include the tables and figures your report discusses.

Assignments that do not conform to the format requirements may not earn credit. After you upload the PDF, assign its pages to the questions in the assignment. If you use an AI tool to convert handwritten work into LaTeX, check the result against your original before you submit. We will grade the PDF you submit, so an error introduced in the conversion will count like any other error.

The notebook. Upload your Jupyter notebook as an `.ipynb` file, with the output cells saved from the run that produced the numbers in your report. The notebook is not graded, but it is required. We will keep it as a record of your code, and we may run it to check your results. A computational problem submitted without its notebook will earn at most 1/3.

Free extensions

You have **1 free homework extension** for the quarter extending a single homework deadline by 24 hours with no questions asked. To use one, email me **before** the original deadline. Beyond this extension, late homework will not be accepted.

F.3 Standards assessments

Rather than traditional midterms and a written final, we will have **four in-class standards assessments**, each 50 minutes long.

Assessment schedule

- **Standards Assessment 1.** Week 3 Wednesday (October 14).
- **Standards Assessment 2.** Week 5 Friday (October 30).
- **Standards Assessment 3.** Week 7 Friday (November 13).
- **Standards Assessment 4.** Week 9 Friday (December 4).

*A minor error is one that still allows you to demonstrate your relative understanding of the problem as intended.

Each standards assessment serves double duty: it introduces **new** standards and provides **retesting** opportunities for all previously assessed standards. If you initially struggled with a particular standard, you have the time and opportunity to study, practice, and try again.

Grading of standards

Each standard is graded on a three-level scale:

- **M** (Meets Expectations): The response is correct, complete, and clearly written, and meets the mastery criterion for that standard. The criterion will be posted when the standard is released.
- **R** (Revision Needed): The work contains a minor error but demonstrates full conceptual understanding. You have the opportunity to come to office hours within **3 business days** of receiving the graded paper, bring your correction already written out, and explain the error and how you fixed it. If the explanation convinces me that you understand the error and how to fix it, the R is upgraded to M. Otherwise, it becomes N.
- **N** (Not Yet): The work contains significant errors, circular reasoning, or does not engage with the core idea. The standard stays available and you can attempt it again at any later sitting.

When you attempt a standard, I will evaluate your work against the mastery criterion for that standard. If your work meets the criterion, the standard is marked M. Otherwise, you will get helpful feedback and the chance to reflect on the feedback, revise your work, and then reassess your understanding.

This feedback loop represents and supports the way that people learn. Learning happens over time, as we revisit ideas and reflect on them. In this class, your final grade will reflect how well you **eventually** understand each topic. You can make mistakes on a standard without penalty, as long as you **eventually** demonstrate fluency in the topic.

Once a standard is marked M, it stays at M permanently. If you have already earned M on a standard, you do not need to attempt it again. I will maintain a record of your **completed** standards (those with M) and periodically update Canvas.

My hope is that this method of grading will keep you clearly informed as to the expectations of the class and how well you are meeting them, while also removing the (often distracting) elements of linear grading that use total points.

Missed assessment

If you believe you will miss a standards assessment, inform me at least a week in advance, if possible (barring emergencies outside your control). Depending on the reason, I may allow a make-up. In most cases, there will be no make-up, as you will have the opportunity to demonstrate the standard again at a subsequent sitting.

F.4 Oral final exam

During finals week (December 8–11), you will have a **25–30 minute oral final exam** with me. This is a holistic assessment of your mathematical understanding and does **not** introduce new standards or provide a path to complete standards not yet at M.

The oral final exam may include:

- Explaining a derivation or a proof from the course;
- Carrying out a short computation by hand, such as a Newton step;
- Reconstructing any homework solution you submitted, or explaining a computational problem you submitted and what its output showed;
- Connecting results across modules, for instance the way a particular theorem reappears in a later module;
- Answering questions on topics that the course covered but did not assess as a standard.

The oral final exam is graded holistically as **Strong**, **Satisfactory**, or **Insufficient**, and adjusts your base grade as described in section G.2. More details on the content of the oral final exam, and on what distinguishes a Strong, a Satisfactory, and an Insufficient performance, will be provided around week 8.

§G. Assessment and grading

Your course grade is calculated in two stages. First, your standards and homework give a **base grade**. Second, your oral final exam adjusts the base grade by plus/minus steps to give your **final grade**. Unexcused absences can limit how far the oral final exam raises your grade.

G.1 Stage 1: Base grade

Your **base grade** is determined by a **checklist**, not a weighted average. To earn a given grade, you must meet **both** requirements in its row. Your base grade is the best letter grade whose row you meet in full.

Grade	Standards at M (out of 10)	Homework score
A	10	At least 90%
B	9	At least 85%
C	8	At least 80%
D	7	At least 75%

Note: If you miss the **D** requirement in either category, the grade is an **F**.

G.2 Stage 2: The oral final exam, attendance, and plus/minus

Your oral final exam can raise your base grade by up to two steps or lower it by one step. A step is one plus-or-minus increment, such as **B** to **B**+. Only base grades of **A**, **B**, and **C** are adjusted. A base grade of **D** or **F** is also your final grade.

- A **Strong** oral final exam raises the base grade by one or two steps. A base grade of **B** can become a **B**+ or an **A**−. With more than two unexcused absences, the raise is limited to one step.
- A **Satisfactory** oral final exam leaves the base grade where it is, or raises it one step if you are close to meeting the requirements for the next letter grade and you have at most two unexcused absences.
- An **Insufficient** oral final exam lowers the base grade by one step. A base grade of **B** becomes a **B**−.

Note: UChicago does not award an **A**+ or a **D**−.

Please contact me at any time during the quarter if you have questions about the grading system or would like to review your current progress.

The final decision regarding any changes to these guidelines will be made by the Director and Co-Directors of Undergraduate Studies in the Department of Mathematics and will be communicated to all via Canvas. Any such changes can only (if anything) loosen the requirements from what is given above.

§H. Collaboration policy

Collaboration and cooperation are extremely helpful in the learning process. However, it is often unclear what exactly "collaboration" means when working on assignments. The following section should clarify my expectations and give guidelines for avoiding plagiarism.

- **Homework:** On weekly homework problems, you are permitted to discuss big ideas and hints with your classmates, but every step of every solution should be one that you understand yourself and that you have generated on your own. On computational problems, you may also discuss code with classmates, but the written report must be your own.

  Any collaboration should occur only when your collaborator is at essentially the same stage of the problem solution as you. In particular, if you have not yet started problem #4 and you ask a friend (who has already completed it), "How did you do problem 4?", this counts as **a violation of the university's academic integrity policy**. To ensure productive collaborations,

do not work in teams larger than three people on any problem at any given time. In teams larger than that, you run the risk of unequal contribution or social loafing.

On your written homework, you must indicate who your collaborators are. (If you collaborate with different people on different problems, just say so!)

- **Assessments:** The standards assessments and the oral final exam are individual. Collaboration, outside resources, and technology of any kind are not permitted.
- **Outside resources:** Unless explicitly permitted, copying outside materials such as solutions found online, web pages, videos, etc., is considered **a violation of the university's academic integrity policy**. This includes library and online implementations of a method you have been asked to implement yourself. So is basing your work on the efforts of another student who previously completed this course or one like it, or a tutor not specifically approved by me.
- **Tools and software:** Unless otherwise specified, you may use a calculator or mathematical software (such as WolframAlpha or a Python session) to check your work without attribution. You will need to include all steps and details in your submitted work.
- **Generative AI:** Generative AI tools are covered separately in section E. Where that section permits something, it takes precedence over this policy.

H.1 Consequences of academic dishonesty

Violations of academic integrity are serious and will be handled seriously. Where an instructor has probable cause that cheating or plagiarism has occurred, the resulting punishment could include (at least) a zero on the homework assignment. On a standards assessment, it could include a permanent N on every standard attempted at that sitting, so that those standards cannot be completed this quarter. For more details regarding academic honesty within the College, please visit the following link: https://college.uchicago.edu/advising/academic-integrity-student-conduct.

H.2 A positive note

Remember that I want you to be successful. That is, I want you to develop a deep, personal understanding of the material we study so that you become a better student of mathematics who can go on to do well in all of your future endeavors. Every part of this course structure, including both collaborative work and restrictions on collaboration, is intended to help you with this.

In all aspects of the course, please understand that I am generous with hints and am always willing to discuss problems with you. I will never simply give you an answer, but I will offer direction and guidance that will assist you in coming up with a solution on your own. This is by far the most satisfying way to solve a problem, and the difficulty is well worth it. You are always welcome to discuss your questions or concerns with me at any time.

§I. Classroom norms

I.1 'Growth', not 'ability'

There is a very prevalent belief that you are either "good" or "bad" at math, and if you are "bad" at it, then you will always be bad at it no matter how hard you try. This is extremely false, and the mathematics community bears a lot of responsibility for perpetuating this myth. In reality, mathematics is just like any other discipline or skill: you can improve more and more with practice.

We are all capable of growth in mathematics. You should measure your success in this class by how much your understanding of the concepts has improved over the course of the quarter. Also, math is very hard, so you should expect to struggle with the material! When you struggle, you are learning and growing. Not all people show their struggles in equal ways, so you should always be wary of judging your progress based on your perception of your peers' struggles. You are probably doing better than you think.

I.2 Respecting each other

We are not all coming to this class with the same privileges, resources, time, and knowledge. It's really important to keep this in mind when working with each other on homework assignments and during class meetings. **It is our strong belief that as a community, mathematicians and scientists need to do a much better job of making our disciplines more accessible to people of all races, genders (including gender non-conforming folks), sexual identities, and class backgrounds.** While this is a priority for us in the classroom, we do not claim to know how to best honor this commitment, but we are eager to listen, adapt, and learn. So **we are very open to feedback from students when it comes to making the course more accessible and inclusive to all identities.**

It's also important to think about how to respect one another when working together in groups. It's not equally easy for all of us to speak up in a large group, and the voices of historically underrepresented/marginalized students are most easily drowned out in group work. So please keep this in mind when working together. Here are some concrete examples of positive collaborative behavior:

- Making sure everyone who wants it has the opportunity to speak frequently. This can mean checking in with each other to make sure everyone is following along and contributing when they have an idea.
- Respecting people's pronouns and other aspects of their identity.
- Making sure that everyone's ideas are acknowledged when writing up the final solution to a problem. When working in groups, solutions often evolve organically; an idea might pop into your head, and you may think it's yours and yours alone, but perhaps you only arrived there because of something else that someone already said. Pay attention to what people are saying and try to learn from one another.

We will do our best to check in with folks periodically during the quarter. If at any time in the quarter, you find yourself in a group of students for which the above behaviors aren't being practiced and people aren't feeling respected, please let me know.

§J. Helpful resources

J.1 Office hours

Office hours are really Student hours! They are an opportunity for you to stop by your instructor's office and ask questions. I have specifically set this time aside in order to give personalized help to individual students. Any and all questions are welcome, whether you are working on a homework question, have questions on a broader concept discussed in class, or potentially even questions beyond the scope of the course.

Typically, students' misunderstandings in mathematics are initially small, but due to the cumulative nature of math, a small misunderstanding can compound as concepts are utilized together. Students who frequently attend office hours and actively engage by asking questions are able to continually resolve their small misunderstandings, resulting in sustained improvement.

Email responses

I do my best to reply to emails promptly and helpfully. However, I receive a lot of email. To help both you and me, here are some specific expectations about emails:

- If you email me between 8:00 AM and 5:00 PM on a weekday, I'll try to reply to you on the same day.
- If you email me in the evening or overnight (after 5:00 PM), I will try to reply to you the next weekday.
- If your email asks a question that is answered in the Syllabus or on Canvas (such as in an announcement or an assignment sheet), I may reply by directing you to read the appropriate document.
- Often, it's much easier to discuss questions in person. I may ask you to meet with me in my office rather than answering directly in an email.

J.2 Academic accommodations

If you require any special academic accommodations, please provide me with a copy of your Accommodation Determination Letter (issued by the Student Disability Services office) as soon as possible, so we can discuss how your accommodations may be

implemented in this course. If you are in the process of obtaining accommodations, please inform me as soon as possible. More information can be found here: https://disabilities.uchicago.edu/.

J.3 Religious accommodations

The University of Chicago is home to students of all the world's major religions and, though firmly a secular institution, values the rich diversity of spiritual expression and practice found on campus. It is therefore the policy of the University that students who miss class, assignments, or exams to observe a religious holiday must be accommodated as follows: (i) absences may not be counted as a missed class in any course in which attendance is a measure of academic performance; (ii) reasonable extensions of time must be given, without academic penalty, for missed assignments; and (iii) exams must be reasonably rescheduled without academic penalty. You must inform me in writing of your need to observe a religious holiday reasonably well in advance of the absence, preferably at the beginning of the quarter. More information can be found at the following: https://provost.uchicago.edu/handbook/clause/policy-religious-accommodation-missed-classes-assignments-and-exams.

J.4 Wellness resources

UChicago has counseling available both 24/7 and by appointment through http://wellness.uchicago.edu. Additionally, you can access medical care, including 24/7 support from medical professionals to address your healthcare questions.

J.5 Sexual misconduct policy

The University of Chicago recognizes that members of the university community are responsible for ensuring that the community is free from discrimination and other forms of sexual misconduct based on sex or gender, including sexual harassment, sexual assault, stalking, domestic violence, and dating violence. Faculty are considered "Individuals with Title IX Reporting Responsibilities" of the University and are obligated to report information to the Title IX Coordinator related to sexual misconduct. If you think your rights, or the rights of someone else in the university community, have been violated, you can find information on resources and reporting at: https://cares.uchicago.edu/.

Title IX Coordinator: Bridget Collier, Associate Provost for Equal Opportunity Programs (bcollier@uchicago.edu, 7737025671)

§K. Recording and deletion policy

The Recording and Deletion Policies for the current academic year can be found in the Student Manual under Petitions, Audio & Video Recording on Campus.

- Do not record, share, or disseminate any course sessions, videos, transcripts, audio, or chats.
- Do not share links for the course with those not currently enrolled.
- Any Zoom cloud recordings will be automatically deleted 90 days after the completion of the recording.

§L. Disclaimer

I reserve the right to make changes to this syllabus if needed. Any changes will be announced to the class in a timely manner.